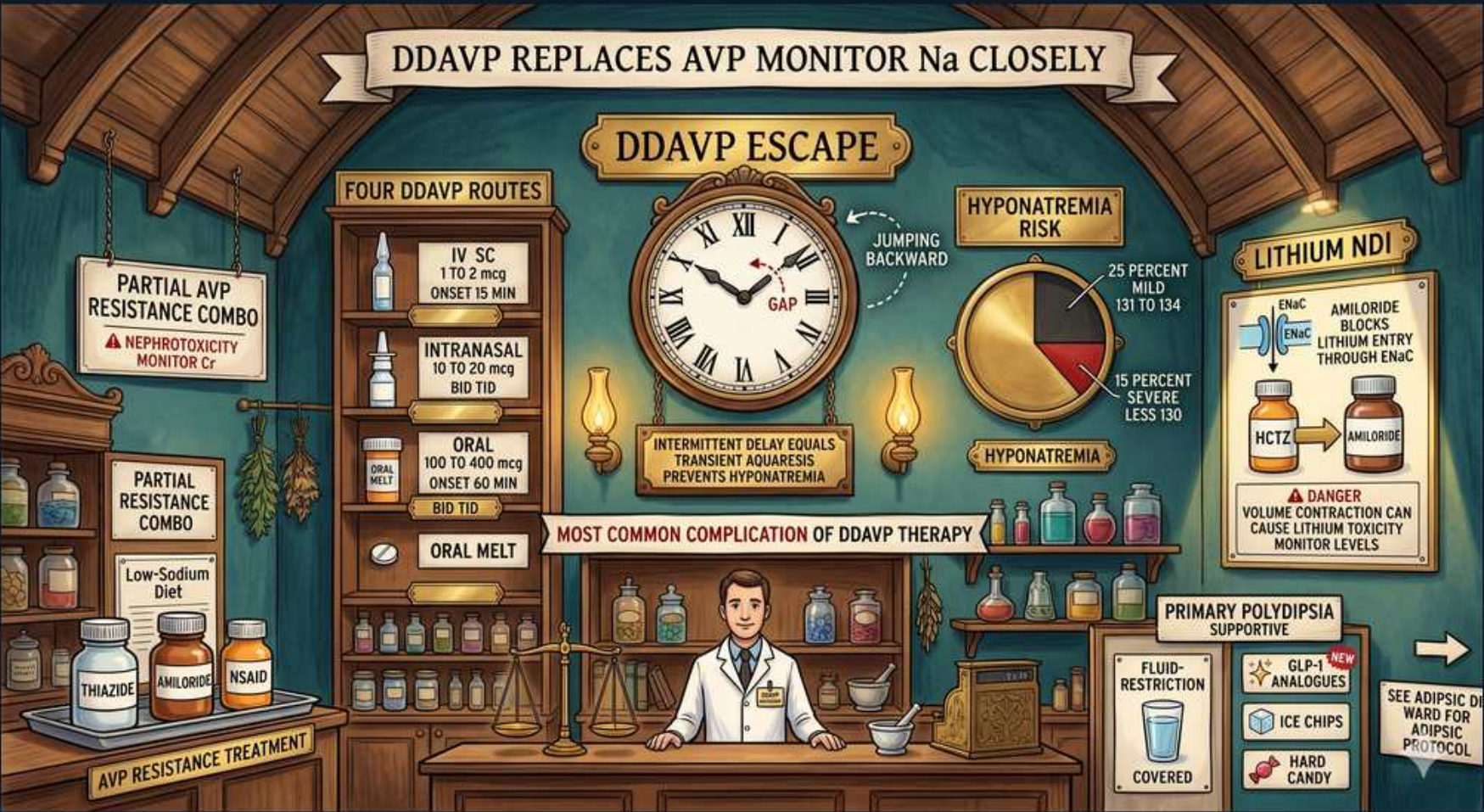


Diabetes insipidus treatment — DDAVP routes and nephrogenic therapy



1. DDAVP replaces AVP

WHAT YOU SEE

Banner: DDAVP replaces AVP, monitor Na closely

WHAT IT ENCODES

Desmopressin is the cornerstone treatment for central DI, replacing deficient AVP and concentrating urine via V2 receptors. Sodium must be monitored closely because over-replacement causes hyponatremia. It is ineffective in nephrogenic DI.

BOARD TRAP

DDAVP treats central/gestational DI; lack of response indicates nephrogenic DI, where you instead use diet, thiazides, and amiloride.

2. IV / SC route

WHAT YOU SEE

IV SC 1-2 mcg, onset 15 min

WHAT IT ENCODES

Parenteral DDAVP (IV/SC) at 1-2 mcg has the fastest onset (~15 min) and highest potency, used acutely or perioperatively. Useful when enteral absorption is unreliable. Requires careful titration to avoid hyponatremia.

BOARD TRAP

Parenteral DDAVP is far more potent than oral — dose conversion errors cause iatrogenic water retention and hyponatremia.

3. Intranasal route

WHAT YOU SEE

Intranasal 10-20 mcg, BID-TID

WHAT IT ENCODES

Intranasal DDAVP (10-20 mcg) is a common maintenance route with intermediate potency. Absorption varies with mucosal congestion or rhinitis. Dosed once to twice daily.

BOARD TRAP

Nasal congestion/URIs reduce intranasal DDAVP absorption, mimicking breakthrough DI — consider absorption before escalating dose.

4. Oral DDAVP

WHAT YOU SEE

Oral 100-400 mcg, onset 60 min, BID-TID

WHAT IT ENCODES

Oral DDAVP (100-400 mcg) has the lowest bioavailability and slower onset (~60 min) but is convenient for chronic therapy. Higher milligram doses reflect poor absorption. Titrated to symptom control.

BOARD TRAP

Oral DDAVP needs much higher mcg doses than nasal/parenteral due to low bioavailability — not a sign of resistance.

5. Oral melt (sublingual)

WHAT YOU SEE

Oral melt formulation listed

WHAT IT ENCODES

Sublingual melt formulations offer more consistent absorption than oral tablets and avoid food interactions. They are convenient for nocturnal polyuria dosing. Still require sodium monitoring.

BOARD TRAP

Food reduces oral tablet DDAVP absorption; the melt is taken away from meals for reliable effect.

6. Intermittent dosing / aquaresis

WHAT YOU SEE

Intermittent delay equals transient aquaresis prevents hyponatremia

WHAT IT ENCODES

A deliberate daily window of breakthrough diuresis (aquaresis) lets retained free water escape, preventing chronic hyponatremia. Patients are taught to delay a dose periodically. This is a key safety strategy in DDAVP therapy.

BOARD TRAP

Continuous DDAVP without an aquaresis window leads to cumulative water retention and hyponatremia — schedule a breakthrough.

7. DDAVP escape / hyponatremia risk

WHAT YOU SEE

Clock jumping backward, hyponatremia risk pie chart

WHAT IT ENCODES

Hyponatremia is the most common complication of DDAVP therapy. Roughly 25% have mild (Na 131-134) and ~15% severe (Na <130) episodes. Risk rises with fixed dosing plus free access to water.

BOARD TRAP

The leading DDAVP complication is hyponatremia, not hypernatremia — counsel against excess fluid plus scheduled drug.

8. Lithium NDI — amiloride

WHAT YOU SEE

Lithium NDI panel, amiloride blocks ENaC

WHAT IT ENCODES

For lithium-induced nephrogenic DI, amiloride blocks ENaC and prevents lithium entry into principal cells. It is the targeted therapy and may allow continued lithium use. Monitor potassium and lithium levels.

BOARD TRAP

Amiloride is preferred for lithium NDI because it blocks lithium uptake; thiazides here can RAISE lithium levels and cause toxicity.

9. HCTZ / thiazide for NDI

WHAT YOU SEE

HCTZ listed in lithium NDI panel

WHAT IT ENCODES

Thiazides paradoxically reduce urine output in nephrogenic DI by inducing mild volume contraction and enhancing proximal sodium/water reabsorption. Combined with a low-sodium diet they are first-line for NDI. Watch for hypokalemia.

BOARD TRAP

Thiazide volume contraction can raise lithium levels — in lithium NDI prefer amiloride or monitor lithium closely if thiazide used.

10. Volume contraction caution

WHAT YOU SEE

Danger: volume contraction can cause lithium toxicity

WHAT IT ENCODES

The thiazide mechanism relies on volume contraction, which reduces lithium clearance and risks lithium toxicity. This makes drug interactions central in lithium NDI management. Sodium restriction augments the antidiuretic effect.

BOARD TRAP

Thiazide-induced volume contraction is therapeutic for NDI but dangerous with concurrent lithium — check lithium and renal function.

11. Partial AVP resistance combo

WHAT YOU SEE

Partial resistance combo: thiazide, amiloride, NSAID, low-sodium diet

WHAT IT ENCODES

Partial nephrogenic DI is managed with combined low-sodium diet, thiazide, amiloride, and an NSAID. NSAIDs reduce urine output by inhibiting prostaglandin-mediated antagonism of AVP. Combination therapy targets multiple steps.

BOARD TRAP

NSAIDs help NDI by blocking prostaglandins that oppose AVP — but watch renal function and GI/cardiac risk with chronic use.

12. Primary polydipsia — supportive

WHAT YOU SEE

Primary polydipsia supportive: fluid restriction, ice chips, hard candy

WHAT IT ENCODES

Primary polydipsia is treated with behavioral fluid restriction and measures like ice chips or hard candy for dry mouth. DDAVP is avoided because the patient keeps drinking, risking hyponatremia. Address underlying psychiatric or dipsogenic cause.

BOARD TRAP

Do NOT give DDAVP for primary polydipsia — combined with continued drinking it causes severe dilutional hyponatremia.

13. Monitor sodium / creatinine

WHAT YOU SEE

Partial AVP resistance combo, monitor Cr

WHAT IT ENCODES

All DI therapy requires monitoring of sodium and renal function. Rapid correction of chronic hypernatremia risks cerebral edema, so correct slowly. Track for both over- and under-treatment.

BOARD TRAP

Correct chronic hypernatremia slowly (about 10 mEq/L per day) to avoid cerebral edema, just as you correct hyponatremia cautiously.
